369) $a = \frac{2}{3}$
$b = 2$
$c = 4$
$d = 2$

368) a) $\log_{81} 9 = x$
$3^4 = 9$
$3^4 = 3^x$
$\boxed{x = 2}$

b) $\log_{2} 128 = x$
$2^x = 128$
$2^x = 2^7$
$\boxed{x = 7}$

$128 \,|\, 2$
$64 \,|\, 2$
$32 \,|\, 2$
$16 \,|\, 2$
$8 \,|\, 2$
$4 \,|\, 2$
$2 \,|\, 2$
1
2^7

370) $a = 49$
$b = 512$
$c = 0{,}2$

b) $\log_{4} 32 = x$
$\left(\frac{1}{2}\right)^2 = 32$
$\left(2^{-1}\right)^2 = 2^5$
$-2x = 5$
$\boxed{x = -5/2}$

$\left(\frac{1}{2}\right) = 2^{-1}$
$\left(\frac{1}{2}\right)^{-1} = \frac{1}{2} = 2^1$

371) $a = 3/2$
$b = 243$
$c = 0$

c) $\log_{\sqrt{3}} 49 = x$
$\left(\sqrt{3}\right)^x = 49$
$\left(3^{\frac{1}{2}}\right)^x = 7^2$
$3^{\frac{x}{2}} = 7^2$
$\frac{x}{2} = 2 \Rightarrow \boxed{x = 4}$

372) $y = \log_{\sqrt{2}} \sqrt{2} + \log_{\sqrt[4]{100}} \sqrt[4]{10} - \log_{0} 100$

$y = \frac{a}{c} + \frac{b}{d} - \frac{c}{f} \Rightarrow y = \frac{3 \cdot 1 - 8}{4} = -\frac{5}{4} = -\frac{3}{2}$

$\log_{\sqrt{2}} \sqrt{2} = a$
$\left(\sqrt{2}\right)^a = \sqrt{2}$
$2^{\frac{a}{2}} = 2^{\frac{1}{2}}$
$\frac{a}{2} = \frac{1}{2}$
$\boxed{a = 1}$

$\log_{\sqrt[4]{100}} \sqrt[4]{0{,}1} = b$
$\left(\sqrt[4]{100}\right)^b = \sqrt[4]{0{,}1}$
$\left(\sqrt[4]{10^2}\right)^b = \sqrt[4]{10^{-1}} \Rightarrow \left(\sqrt{10}\right)^b = \sqrt[4]{10^{-1}}$
$10^{\frac{b}{4}} = 10^{-\frac{1}{4}}$
$\frac{b}{4} = -\frac{1}{4}$
$\boxed{b = -1}$

$\log_{0} 100 = c$
$10^c = 100$
$10^c = 10^2$
$\boxed{c = 2}$

373) $c < d < b < a$.

$\log_{\sqrt{2}} \sqrt{2} = a$ $\quad \log_{0{,}001} 0{,}001 = b$

$\log_{0{,}25} 0{,}5 = c$ $\quad \log_{100} 0{,}001 = d$

374) $y = 3/2$

$y = \log_{\sqrt{3}} (\log_3 9) + \log_{\sqrt{2}} (\log_2 4)$

$\log_3 9 = x$
$3^x = 9$
$3^x = 3^2$
$\boxed{x = 2}$

$\log_2 4 = x$
$2^x = 4$
$2^x = 2^2$
$\boxed{x = 2}$

$y = \log_{\sqrt{3}} 2 + \log_{\sqrt{2}} 3$

$y = \frac{1}{2} + \frac{1}{2} \Rightarrow y = \frac{1+1}{2}$

$\boxed{y = 3/2}$

$\log_{10} 1000 = x$
$10^x = 1000$
$10^x = 10^3$
$\boxed{x = 3}$

$\log_{4} 2 = x$
$4^x = 2$
$2^{2x} = 2^1$
$\boxed{x = 1/2}$

375) $x = \frac{-20}{9}$

$\log_{\sqrt[3]{81}} \sqrt[3]{\sqrt{3}} = x \Rightarrow \left(\frac{1}{\sqrt[3]{81}}\right)^x = \sqrt[3]{\sqrt{3}}$

$\left(\frac{1}{3^{\frac{4}{3}}}\right)^x = 3^{\frac{1}{6}}$

$3^{-\frac{4}{3}x} = 3^{\frac{1}{6}}$

$\sqrt[3]{\sqrt{3}}$

$\sqrt[3]{\sqrt{3}} = 3^{\frac{1}{6}}$

$\sqrt[3]{\sqrt{3}} = \sqrt[3]{\sqrt{3}}$

$\frac{1}{\sqrt[3]{3^4}} = 3^{\frac{1}{6}}$
$\frac{1}{3^{\frac{4}{3}}} = \left(3^{\frac{1}{6}}\right)^3$
$\frac{1}{3^{\frac{4}{3}}} = 3^{\frac{1}{2}}$

$81 \,|\, 3$
$27 \,|\, 3$
$9 \,|\, 3$
$3 \,|\, 3$
1

$\Rightarrow 3^{\frac{1}{6}}$

$-\frac{4}{3}x = \frac{1}{2}$
$-8x = 20$
$\boxed{x = -\frac{20}{9}}$

376) a) $x = 1/2$
b) $x = -2$

a) $\log_{\sqrt{m}} x = 2$
$\left(m^{\frac{1}{2}}\right)^2 = \sqrt[3]{m}$
$m^{2x} = m^{\frac{1}{3}}$
$2x = \frac{1}{3}$
$x = \frac{1}{6}$

b) $\log_{\sqrt{m}} \frac{1}{m} = x$
$\left(\sqrt{m}\right)^x = \frac{1}{m}$
$m^{\frac{x}{2}} = m^{-1}$
$\frac{x}{2} = -1$
$\boxed{x = -2}$

$X = \frac{1}{3} \cdot \frac{1}{2} \Rightarrow \boxed{x = 1/6}$

377) a) 2
b) 9
c) 625

a) $3^{\log_3 2} = 2$
b) $9^{\log_9 3}$
$\left(2^{\log_2 3}\right)^2 = (3)^2 = 9$
c) $16^{\log_2 5}$
$\left(2^{\log_2 5}\right)^4 = 5^4 = 625$

378) 3

Propriedades operatórias

1) logaritmo do produto $0 < a \neq 1$, $b > 0$, $c > 0$

$\log_a (b \cdot c) = \log_a b + \log_a c$

$\log_a b = x \Rightarrow a^x = b$
$\log_a c = y \Rightarrow a^y = c$
$\log_a (b \cdot c) = z \quad a^z = bc$
$\Big\} \; a^z = a^x \cdot a^y \Rightarrow z = x + y$

Exemplo

$\log_a (3 \cdot 4) \Rightarrow \log_a 3 + \log_a 4 \Rightarrow \log_a 3 + 2$

$\log_3 (7 \cdot 5 \cdot 6) \Rightarrow \log_3 7 + \log_3 5 + \log_3 2 \Rightarrow$

$\log_{10} 210 \Rightarrow \log_{10} 2 + \log_{10} 3 + \log_{10} 5 + \log_{10} 7$

$= 0{,}3 + 0{,}47 + 0{,}7 + 0{,}85 = 2{,}32$

$210 \,|\, 2$
$105 \,|\, 3$
$35 \,|\, 5$
$7 \,|\, 7$
1

$210 = 2 \cdot 3 \cdot 5 \cdot 7$

logaritmo do quociente

$\log_a \frac{b}{c} \Rightarrow \log_a b - \log_a c$

$\log_a b = x \Rightarrow a^x = b$
$\log_a c = y \Rightarrow a^y = c$
$\log_a \frac{b}{c} = z \Rightarrow a^z = \frac{b}{c}$
$\Big\} \; a^z = \frac{a^x}{a^y} \Rightarrow z = x - y$

Exemplo

$\log_2 \frac{3}{4} \Rightarrow \log_2 3 - \log_2 4 \Rightarrow \log_2 3 - 2$

$\log_3 \frac{4}{7} \Rightarrow \log_3 4 - \log_3 7$

$\log_{10} \frac{6}{5} \Rightarrow \log 6 - \log 5 \Rightarrow (\log 2 + \log 3) - \log 5$

$\log 2 = 0{,}3$
$\log 3 = 0{,}47$
$\log 5 = 0{,}7$

$(0{,}3 + 0{,}47) - 0{,}7 = 0{,}07$

logaritmo da Potência

$r \in \mathbb{R}$
$b > 0$
$0 < a \neq 1$

$\log_a b^r \Rightarrow r \cdot \log_a b$

$\log_a b = x \Rightarrow a^x = b$
$\log_a b^r = y \Rightarrow a^y = b^r$
$\Big\} \; a^y = (a^x)^r \Rightarrow y = x \cdot r$

Exemplo

$\log_2 3^7 \Rightarrow 7 \log_2 3$

$\log_3 \sqrt{5} \Rightarrow \log_3 5^{\frac{1}{2}} \Rightarrow \frac{1}{2} \log_3 5$

$\log_3 \frac{1}{16} \Rightarrow \log_3 2^{-4} \Rightarrow -4 \log_3 2$

1) $\log_a (b \cdot c) \Rightarrow \log_a b + \log_a c$

2) $\log_a \frac{b}{c} \Rightarrow \log_a b - \log_a c$

3) $\log_a b^r \Rightarrow r \log_a b$

$0 < a \neq 1$
$b > 0$
$c > 0$
$r \in \mathbb{R}$

Exemplo

$\log_a \frac{x^2}{\sqrt[3]{y}}$ $\quad \log_a x = 2$ $\quad 2 \log_a y = 3$

$\log_a \frac{x^2}{\sqrt[3]{y}} \Rightarrow \log_a x^2 - \log_a \sqrt[3]{y}$

$2 \log_a x - \log_a y^{\frac{1}{3}}$

$2 \log_a x - \frac{1}{3} \log_a y$

$2 \cdot (-2) - \frac{1}{3} \cdot 3$

$-4 - 1 = -5$

Exemplo 2

Se $\log E = 4 \cdot \log a + 2 \log b - \log c$, determine E

$\log E = \underbrace{\log 10^4 + \log a^4 + \log b^2}_{} - \log c$

$\log E = \log \left(\frac{10^4 a^4 b^2}{c}\right)$

$\boxed{E = \frac{10^4 a^4 b^2}{c}}$